

drain, as will be soon seen, is $\frac{1}{g_{m2}} || r_{o1} || r_{o2}$. In most cases, however, r_o will be much greater than $\frac{1}{g_m}$, and the approximation is an accurate one.

Example 21.1

Determine the small-signal AC gain of the circuit shown in Fig. 21.3 using the intuitive method presented in Eq. (21.1).

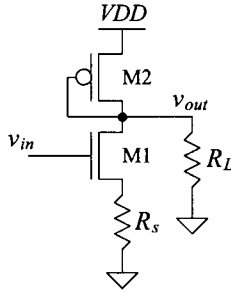


Figure 21.3 Amplifier analyzed in Ex. 21.1.

The small-signal gain of this circuit is given by

$$A_v = \frac{v_{out}}{v_{in}} = - \frac{\text{resistance in the drain}}{\text{resistance in source}}$$

The resistance in the drain is the parallel combination of all resistances connected to the drain of M1, given by

$$\text{Resistance in the drain} = \frac{1}{g_{m2}} || R_L \text{ for } r_{o1} || r_{o2} \gg \frac{1}{g_{m2}}$$

The resistance in the source is the sum of the resistance looking into the source of M1 ($1/g_{m1}$) and the resistance connected from the source of M1 to ground. For the present problem, this resistance is given by

$$R_{\text{in the source}} = \frac{1}{g_{m1}} + R_s$$

The voltage gain is then given by

$$A_v = - \frac{\frac{1}{g_{m2}} || R_L}{\frac{1}{g_{m1}} + R_s} \text{ which for } R_L \rightarrow \infty \text{ and } R_s \rightarrow 0 \text{ reduces to } - \frac{g_{m1}}{g_{m2}} \blacksquare$$

We can determine the exact gain of the amplifier of Fig. 21.1a using the full small-signal model shown in Fig. 21.4. Using KCL at the output of the amplifier gives

$$g_{m1} v_{in} + \frac{v_{out}}{r_{o2} || r_{o1}} = -g_{m2} v_{out} \quad (21.2)$$

or

$$A_v = \frac{v_{out}}{v_{in}} = - \frac{g_{m1}}{g_{m2} + \frac{1}{r_{o1} || r_{o2}}} = - \frac{\frac{1}{g_{m2}} || r_{o1} || r_{o2}}{\frac{1}{g_{m1}}} = - \frac{\text{resistance in the drain}}{\text{resistance in the source}} \quad (21.3)$$

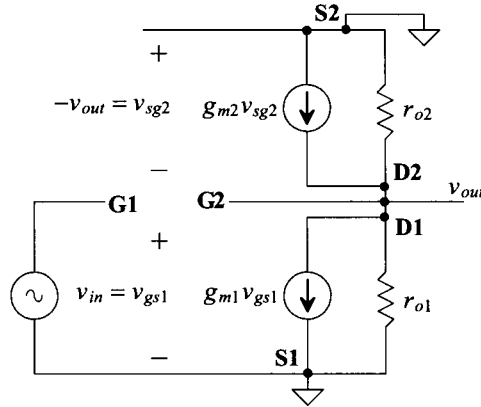


Figure 21.4 Small-signal model of the amplifier shown in Fig. 21.1(a).

When $\frac{1}{g_{m2}} \ll r_{o1} \parallel r_{o2}$, this reduces to

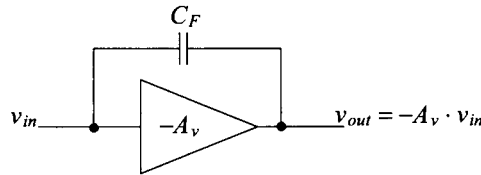
$$A_v = -\frac{\frac{1}{g_{m2}}}{\frac{1}{g_{m1}}} = -\frac{g_{m1}}{g_{m2}} \quad (21.4)$$

which is the same form as Eq. (21.1).

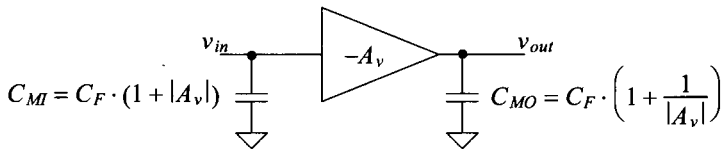
Miller's Theorem

Consider the inverting amplifier seen in Fig. 21.5a with a gain of $-A_v$ and a feedback capacitance between the output and the input of the amplifier. The current supplied by v_{in} or v_{out} to the capacitor is

$$i = \frac{v_{in} - v_{out}}{1/j\omega C_F} = j\omega C_F \cdot (1 + |A_v|) \cdot v_{in} = j\omega C_F \cdot \left(1 + \frac{1}{|A_v|}\right) \cdot v_{out} \quad (21.5)$$



(a)



(b)

Figure 21.5 The derivation of Miller's theorem.

In Fig. 21.5b we replace the feedback capacitor with capacitors on the input and the output of the amplifier to ground. These (adjusted size) capacitors require the same currents from the input source and the output of the amplifier. Note that this substitution, Eq. (21.5), is often called *Miller's theorem*, and the feedback capacitor, C_F , is called the *Miller capacitance*.

Notice that the effective capacitance on the input of the amplifier is multiplied by the gain of the amplifier. For big gains, $|-A_v|$, this means that the input capacitance of the amplifier can be large and can result in slow circuits. Intuitively, this can be understood by realizing that the effective voltage across C_F is $A_v + 1$ times larger than the input signal itself. This large voltage drop across C_F (one side going up by v_{in} while the other side goes down by $A_v \cdot v_{in}$) results in a large displacement current, which makes the capacitor appear larger than it really is.

Frequency Response

Now consider the frequency response of the CS amplifier of Fig. 21.1a. The high-frequency (meaning that the device capacitances are included in the circuit) equivalent circuit is shown in Fig. 21.6a. The gate-drain capacitance of M2 isn't drawn because the gate and drain of M2 are shorted together. A source resistance is added to the circuit to model the effects of the driving source impedance. Miller's theorem is used to break C_{gd1} into two parts: a capacitance at the gate of M1 to ground and a capacitance from the drain of M1 to ground, as seen in Fig. 21.6b (note that the low-frequency gain of the amplifier in Fig. 21.1a is $-g_{m1}/g_{m2}$). Two RC time constants exist in this circuit: one on the input of the circuit and one on the output of the circuit. The input time constant is given by

$$\tau_{in} = R_s(C_{M1} + C_{gs1}) \quad (21.6)$$

where the Miller capacitance at the input, C_{M1} , is

$$C_{M1} = C_{gd1} \left(1 + \frac{g_{m1}}{g_{m2}} \right) \quad (21.7)$$

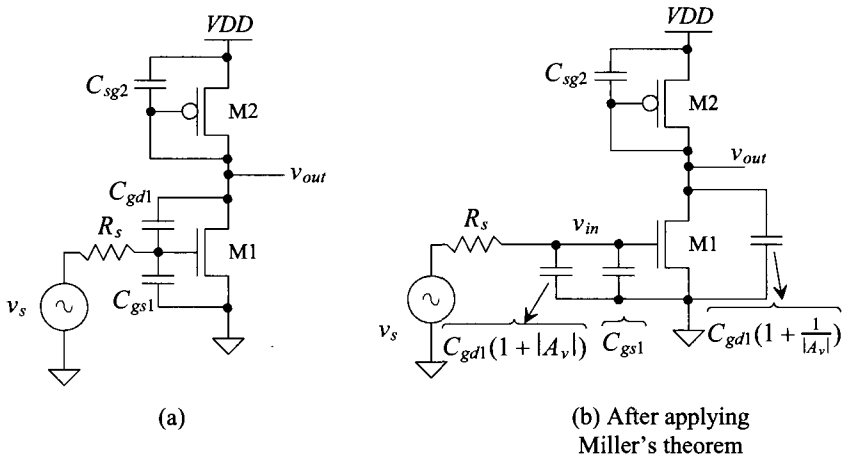


Figure 21.6 Frequency response of the CS amplifier with gate-drain-connected load.

The output time constant is found in a similar way and is given by

$$\tau_{out} = \frac{1}{g_{m2}} \cdot (C_{sg2} + C_{MO}) \quad (21.8)$$

The Miller capacitance which appears at the output, C_{MO} , now becomes

$$C_{MO} = C_{gd1} \left(1 + \frac{g_{m2}}{g_{m1}}\right) \quad (21.9)$$

The frequency response of the amplifier in Fig. 21.1a (Fig. 21.6a) is given by

$$A_v(f) = \frac{-\frac{g_{m1}}{g_{m2}}}{\left(1 + j\frac{f}{f_{in}}\right)\left(1 + j\frac{f}{f_{out}}\right)} \quad (21.10)$$

where the pole associated with the amplifier's input is located at

$$f_{in} = \frac{1}{2\pi\tau_{in}} \quad (21.11)$$

and the pole associated with the amplifier's output is located at

$$f_{out} = \frac{1}{2\pi\tau_{out}} \quad (21.12)$$

The Right-Half Plane Zero

It should be noted that when using Miller's theorem a zero is neglected. If we take a look at Fig. 21.5a, we see that at high frequencies C_f shorts the amplifier's output to its input. Our model in Fig. 21.5b doesn't show this shorting. What this means for the amplifier in Fig. 21.1a or its model in Fig. 21.6a is that C_{gd1} shorts the output to the input at high frequencies.

To characterize this high-frequency effect, where we can't use Miller's theorem, consider the circuit seen in Fig. 21.7. In this figure we model the amplifier in Fig. 21.6 (or Fig. 21.1) with a small-signal resistance on the output of the circuit called $R_o = \frac{1}{g_{m2}} || r_{o2} || r_{o1} \approx \frac{1}{g_{m2}}$ and a capacitance on the output, without including C_{gd1} , of $C_o = C_{sg2}$. We don't concern ourselves with the input time constant but rather just look at the output of the circuit. Summing the currents on the output of the model results in

$$\frac{v_{out} - v_{in}}{1/j\omega C_{gd1}} + \frac{v_{out}}{R_o || 1/j\omega C_o} + g_{m1} \cdot v_{in} = 0 \quad (21.13)$$

Solving for v_{out}/v_{in} gives

$$\frac{v_{out}}{v_{in}} = -g_{m1} R_o \cdot \frac{1 - j\omega \frac{C_{gd1}}{g_{m1}}}{1 + j\omega(C_{gd1} + C_o) \cdot R_o} \quad (21.14)$$

The pole should be recognized as what we wrote in Eq. (21.12) if the gain of the amplifier is small ($A_v = -g_{m1}/g_{m2} \approx 0$). In the numerator we see a right-half plane zero at

$$f_z = \frac{g_{m1}}{2\pi C_{gd1}} \quad (21.15)$$

A zero in the right-half plane (RHP) results in the same magnitude response as a zero in the left-half plane (LHP). However, the phase response is different. A zero in the RHP has the same influence on the phase of a system as a pole in the LHP. Understanding this

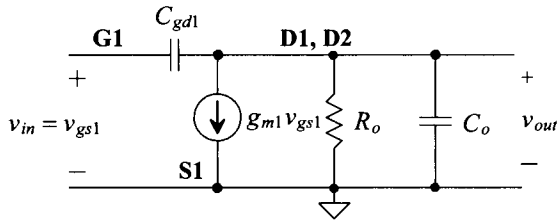


Figure 21.7 Small-signal model used to calculate the RHP zero for amplifiers in Fig. 22.1a–d.

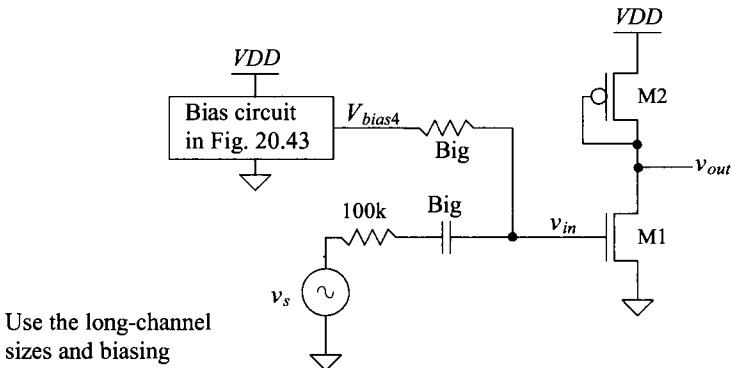
point is important. For the inverting amplifiers (which mean that at low frequencies their phase shift is 180°) in Fig. 21.1, for example, the input feeding directly to the output because of the RHP zero, without the inversion, can result in instability when the amplifier is used with feedback. The output can feed back and add to the input (positive feedback). The result is that the output of the amplifier will grow without limit (and, ultimately, the output of the amplifier will oscillate back and forth between some voltages). The magnitude response of the RHP zero is usually ignored because its value typically places it well beyond the pole frequency. However, in CMOS op-amp design, the phase shift of this RHP zero can, and will, result in stability and/or settling time issues.

Example 21.2

Determine the gain, magnitude, and phase shift of the amplifier shown in Fig. 21.8. Using an AC SPICE simulation, verify your hand calculations.

The bias circuit is used to bias the amplifier at the operating point seen in Table 9.1, that is, at a drain current of $20\ \mu\text{A}$. The big resistor and capacitor set the DC bias point but don't affect the AC operation of the circuit.

$$A_v = -\frac{\frac{1}{g_{m2}}}{\frac{1}{g_{m1}}} = -\frac{g_{m1}}{g_{m2}} = -\frac{150\ \mu\text{A}/V}{150\ \mu\text{A}/V} = -1$$



Use the long-channel sizes and biasing seen in Table 9.1.

Figure 21.8 Amplifier used in Ex. 21.2.

From Table 9.1, $C_{gs1} = 23.3 \text{ fF}$ and $C_{gd1} = 2 \text{ fF}$. The Miller input capacitance, because the gain of the amplifier is -1 , is then 4 fF . The input pole is located at

$$f_{in} = \frac{1}{2\pi \cdot R_s(C_{MI} + C_{gs1})} = \frac{1}{2\pi \cdot 100k \cdot (27.3 \text{ fF})} = 58 \text{ MHz} \quad (21.16)$$

From Table 9.1, $C_{sg2} = 70 \text{ fF}$. The Miller output capacitance is, again, 4 fF . The output pole is located at

$$f_{out} = \frac{1}{2\pi \cdot \frac{1}{g_{m2}} \cdot (C_{MO} + C_{sg2})} = \frac{1}{2\pi \cdot 6.5k \cdot (74 \text{ fF})} = 331 \text{ MHz} \quad (21.17)$$

The zero is located at

$$f_z = \frac{g_{m1}}{2\pi C_{gd1}} = \frac{150 \mu\text{A/V}}{2\pi \cdot 2 \text{ fF}} = 11.9 \text{ GHz}$$

The transfer function of the amplifier is then

$$A_v(f) = \frac{v_{out}(f)}{v_s(f)} = -\frac{\left(1 - j\frac{f}{11.9 \text{ GHz}}\right)}{\left(1 + j\frac{f}{58 \text{ MHz}}\right)\left(1 + j\frac{f}{331 \text{ MHz}}\right)}$$

Figure 21.9 shows the AC analysis simulation results for the amplifier of Fig. 21.8. What would happen if we used v_{in} instead of v_s ? We wouldn't see the input

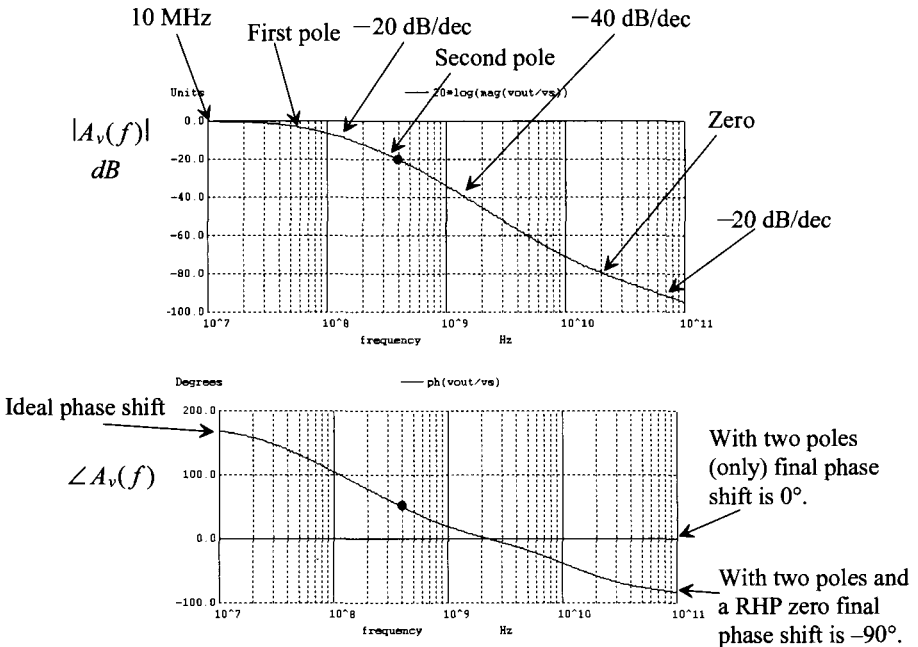


Figure 21.9 Frequency response of the amplifier in Fig. 21.8.

pole's effects on the frequency response. In the magnitude response we see that right after the first pole the gain decreases by a factor of 10 for every increase in frequency by 10 (equivalent to saying -20 dB/decade). We could also say that the gain decreases by 2 for every increase in frequency by 2 (equivalent to saying -6 dB/octave). After the second pole, the gain decreases by 100 for every increase in frequency by 10. The zero causes the frequency response to go back to -20 dB/decade. (The output and input of M1 are shorted together so that all we see are the effects of R_s charging the input capacitance of the amplifier.) It is useful to look at, and understand, the behavior of the amplifier from v_{in} to v_{out} in the simulation.

The effects of the RHP zero occur one decade before f_z . This *excess phase* shift is the problem we discussed earlier. It makes the amplifier appear not like an inverting amplifier (phase shift of -180 degrees) but rather like a noninverting amplifier sooner (at lower frequencies).

Note that magnitude of voltage gains in the neighborhood of 1 are common when using gate-drain-connected loads. At this point it may appear that this amplifier is almost worthless; however, we will see that when this amplifier is used with additional circuitry (e.g., a current source load for a differential pair), it can be very useful. ■

Example 21.3

Determine the magnitude and phase of the time-domain signal v_{out} , at 400 MHz, for the amplifier in Ex. 21.2 based on the frequency plots seen in Fig. 21.9. Verify the results using SPICE.

The magnitude response at 400 MHz is

$$\left| \frac{v_{out}}{v_s} \right| = -20 \text{ dB} = 0.1$$

and the phase response is

$$\angle \frac{v_{out}}{v_{in}} = 50^\circ = -310^\circ$$

The positive value means that the output is leading, or occurring earlier in time, than the input. Remembering that phase shift is simply an indication of a time delay at a particular frequency, we can write

$$50^\circ = 360 \cdot \frac{t_d}{T} = 360 \cdot t_d \cdot 400 \text{ MHz} \rightarrow t_d \approx 350 \text{ ps}$$

for a period of 2.5 ns. We'll use an input voltage of 1 mV to ensure small-signal operation. In the simulation that generated Fig. 21.9 our input AC signal was 1 V. Again, remember that SPICE replaces the active devices with linear models in an AC simulation so any value of amplitude can be used for the input signals. Using 1 V in an AC simulation is convenient because of the ratios often used when calculating gains. Figure 21.10 shows the SPICE simulation results. Notice how the peak amplitude of the input is 1 mV while the peak amplitude of the output (remembering the DC output voltage is $V_{DD} - V_{SG}$ or, from Table 9.1, approximately 3.85 V) is 0.1 mV (-20 db down from the low-frequency value). In

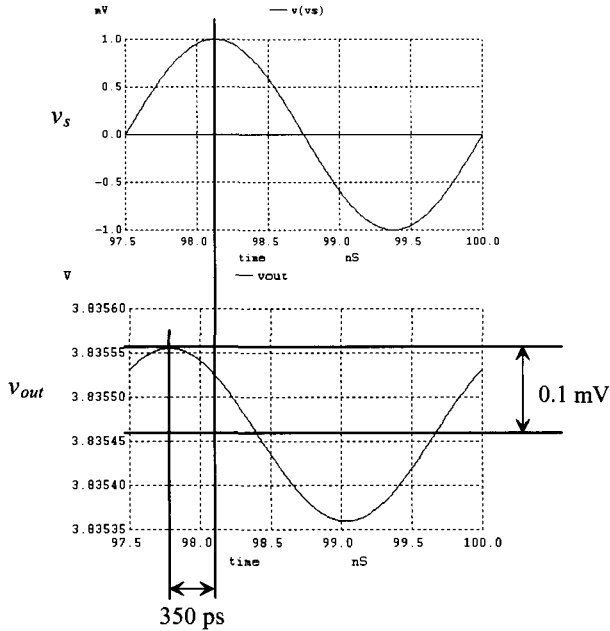


Figure 21.10 Time-domain behavior of the amplifier in Fig. 21.8 at 400 MHz.

a transient simulation we have to wait some time before the circuit settles to a stable operating point. In Fig. 21.10 we've waited 97.5 ns before looking at the output voltage. This gives the bias circuit, for example, time to turn on and generate the bias voltage needed in the amplifier (the value of the big capacitor was also reduced in this transient simulation so that its charging wouldn't increase the simulation time.) ■

A Common-Source Current Amplifier

The previous discussion was centered around CS voltage amplifiers. Next consider the circuit shown in Fig. 21.11. This amplifier can be thought of as a transimpedance amplifier, that is, voltage out and current in. The ideal input impedance of an amplifier with a current input is zero (all of the input current flows into the amplifier with this ideal input impedance). Note that the biasing of the amplifier is controlled by the current mirror formed between M3 and M1 (drawn in a different way to ensure that the reader recognizes the topology when it is seen).

The input impedance of the amplifier in Fig. 21.11 is given by

$$R_{in} = \frac{1}{g_{m3}} || r_{o3} || r_{o4} \approx \frac{1}{g_{m3}} \quad (21.18)$$

The input current, i_{in} , is determined by

$$i_{in} = g_{m3} v_{in} = g_{m3} v_{gs3} = g_{m3} v_{gs1} \quad (21.19)$$

The current through M1, and thus M2, is given by

$$i_d = g_{m1} v_{in} = \frac{g_{m1}}{g_{m3}} \cdot i_{in} \quad (21.20)$$

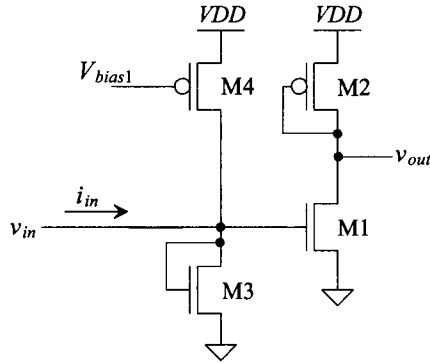


Figure 21.11 A transimpedance amplifier.

This equation can be rewritten, $g_{m3} = \beta_3(V_{GS3} - V_{THN})$ and $g_{m1} = \beta_1(V_{GS3} - V_{THN})$, as

$$\frac{i_d}{i_{in}} = \frac{\beta_1}{\beta_3} = \frac{W_1 L_3}{W_3 L_1} \quad (21.21)$$

That is, we can get a current gain simply by adjusting the size of M1 and M3. The current in M2 can then be mirrored out to a load. The transresistance gain of this configuration is given by

$$A_R = \frac{v_{out}}{i_{in}} = -\frac{i_d \frac{1}{g_{m2}}}{i_d \cdot \frac{g_{m3}}{g_{m1}}} = -\frac{g_{m1}}{g_{m2} g_{m3}} \quad (21.22)$$

Note that adding M3 and thus the resistor of value $\frac{1}{g_{m3}}$ to ground lowers the input time constant and increases the bandwidth of the amplifier.

Common-Source Amplifier with Source Degeneration

The amplifier in Fig. 21.3 is an example of a CS amplifier with source degeneration. The gain of the amplifier is reduced because the effective transconductance of the amplifier becomes

$$\frac{1}{g_{m,eff}} = \frac{1}{g_m} + R_s \quad (21.23)$$

as seen in Ex. 21.1. Adding the source resistance is useful, especially if $R_s \gg 1/g_m$, to precisely set the transconductance of an amplifier. Note that we didn't, in Ex. 21.1, include the resistance looking into the drain of M1 when we wrote the "resistance in the drain." *Let's review, or again verify, how to determine the small-signal, low-frequency resistances looking into the drain and source of a MOSFET.* Using this information will greatly aid in our intuitive analysis of CMOS circuits.

Consider the test circuit shown in Fig. 21.12. Here the DC voltages ensure that the MOSFET is operating in the saturation region. We are concerned with the AC current that flows through v_i . This test voltage divided by i_d gives us the small-signal AC resistance looking into the drain of the MOSFET. In other words, this is the resistance connected to ground at the drain node.

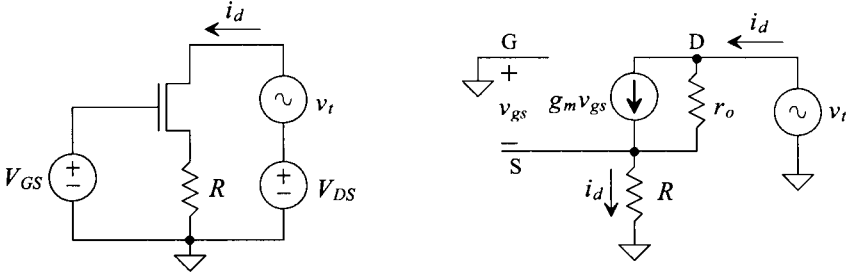


Figure 21.12 Circuit used to determine the resistance looking into the drain of a MOSFET.

Neglecting the body effect, we can write the test voltage as

$$v_t = (i_d - g_m v_{gs})r_o + i_d R \quad (21.24)$$

where (noting $v_g = 0$ and $v_s = i_d R$)

$$v_{gs} = -i_d R \quad (21.25)$$

The resistance looking into the drain of the MOSFET is then given by

$$R_o = r_d = \frac{v_t}{i_d} = (1 + g_m R)r_o + R \approx (1 + g_m R)r_o \quad (21.26)$$

keeping in mind that $r_o \approx \frac{1}{\lambda I_D}$. For a cascoded MOSFET topology ($R = r_o$), this is approximately, as seen in Eq. (20.53), $g_m r_o^2$. For a *triple cascode*, three MOSFETs in series, ($R = g_m r_o^2$), the output resistance is

$$R_o \approx g_m^2 r_o^3 \quad (21.27)$$

noting that the output resistance goes up with each added MOSFET by the open-circuit gain, $g_m r_o$.

Example 21.4

Repeat Ex. 21.1 without neglecting the loading effects of the output resistances of M1 and M2.

The resistance looking into the drain of M2 is $\frac{1}{g_{m2}} || r_{o2}$, while the resistance looking into the drain of M1 is $r_{o1}(1 + g_{m1}R_s)$. The exact gain of the circuit is

$$A_v = \frac{v_{out}}{v_{in}} = - \frac{\frac{1}{g_{m2}} || r_{o2} || [r_{o1}(1 + g_{m1}R_s)] || R_L}{R_s + \frac{1}{g_{m1}}}$$

or when $r_{o1} || r_{o2} \gg \frac{1}{g_{m2}}$, this reduces to

$$A_v = \frac{\frac{1}{g_{m2}} || R_L}{R_s + \frac{1}{g_{m1}}} \blacksquare$$

The resistance in the source of the MOSFET can be found using the circuit shown in Fig. 21.13. Neglecting the body effect and output resistance of the MOSFET gives

$$v_{in} = v_{gs} + i_d R \quad (21.28)$$

Since $g_m v_{gs} = i_d$, this equation can be written as

$$v_{in} = \frac{i_d}{g_m} + i_d R = i_d \left[\frac{1}{g_m} + R \right] \quad (21.29)$$

We can look at this in a simplified manner; that is, $\frac{1}{g_m}$ is the resistance looking into the source of a MOSFET, while R is the resistance connected from the source of the MOSFET to ground.

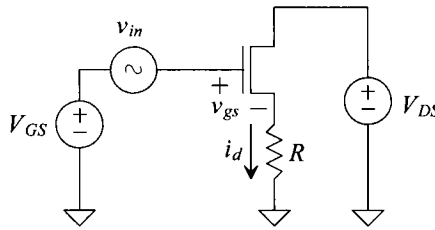


Figure 21.13 Circuit used to determine the resistance looking into the source of a MOSFET.

Noise Performance of the CS Amplifier with Gate-Drain Load

Figure 21.14 shows the CS amplifier of Fig. 21.1a with each MOSFET's model for noise power spectral density included (see Eq. [9.66]). Remembering that we only add the power from each noise source, we can write the output noise power spectral density as

$$V_{onoise}^2(f) = \left(\frac{1}{g_{m2}} \right)^2 \cdot (I_{M1}^2 + I_{M2}^2) \quad (21.30)$$

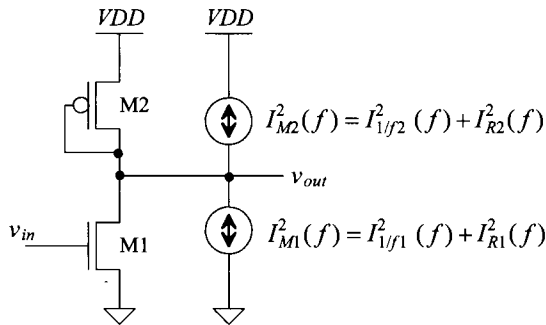


Figure 21.14 Noise modeling in a CS amplifier.